

A Multi-Axis Robustness Analysis of Retrieval-Augmented Generation for Multi-Hop Requirements Traceability

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Abstract. Reported verdicts on GraphRAG versus vector RAG disagree, and the evidence is typically tied to a single corpus, embedder, and judge – and, we show, to where citation quality is measured. Holding a five-pipeline matrix fixed, we vary the embedder (three vendors), the corpus (DO-178C requirements, MuSiQue, 2WikiMultihopQA), the graph traversal, and the judge (GPT-5.4, GPT-4.1, Gemini), over 13,400 generation runs and more than 30,000 faithfulness judgments. GraphRAG’s walk floods the context at precision 0.12–0.23 while its synthesizer cites selectively at 0.48–0.98, so scoring the retrieved set ranks GraphRAG below the vanilla baseline and scoring the answer’s citations ranks it first – in all six settings, under a second traversal, and against two mitigations. Answer-level winners are stratum-conditional but stable across embedders and seeds. Faithfulness declines with hop distance on all three corpora once adequately powered, overturning the corpus-conditional reading that smaller judged subsets suggest; the decline tracks context expansion, with a significant hop×expansion interaction ($p=10^{-7}$). Judges are not stable: on frozen inputs, self-agreement falls to $\kappa \leq 0.14$ after eleven weeks, and two vendors reverse which is stricter between corpora. RAG architecture claims should be tested at this level of robustness, including the citation-measurement point.

Keywords: Retrieval-augmented generation · GraphRAG · Requirements traceability · Multi-hop QA · LLM-as-judge · Evaluation methodology

1 Introduction

Certifying avionics software under DO-178C requires demonstrating *traceability*: each requirement must be linked to the parent it derives from, the system-level intent it satisfies, and the verification artifact that tests it, and certification authorities increasingly require these chains to be auditable across two or more hops [7, 8, 24, 26]. Requirements management tools record those links explicitly, so a requirements corpus is naturally read as a typed link graph, which makes retrieval over it a graph problem as much as a text problem.

RAG architectures for this task have produced contradictory verdicts: Microsoft GraphRAG [6] exploits structure but loses to vector RAG on detailed

queries [15, 31]; Adaptive-RAG [17] routes by complexity yet is hop-blind; LiSSA [10] and TVR [23] omit typed-graph reasoning. Prior evaluations report which architecture wins in a given setting, but not why the outcome changes when the setting changes.

We address this with a **multi-axis robustness** analysis. Holding a five-pipeline architecture matrix fixed (vanilla, agentic, agentic+graph, GraphRAG, learned adaptive), we vary four orthogonal axes: the retrieval embedder (e5-small 384d, Azure 3-small 1536d, bge-m3 1024d – three vendors, three dimensionalities), the corpus (DO-178C typed-edge requirements; Wikipedia paragraph chains via MuSiQue and 2WikiMultihopQA), the graph traversal (a bounded 2-hop walk and personalized PageRank over the same graph), and the faithfulness judge (GPT-5.4, GPT-4.1, Gemini 3.7-flash). The design produces 13,456 generation runs and over 30,000 faithfulness judgments.

Our headline result is not that one architecture wins, but that a methodological choice most evaluations leave implicit – whether citation quality is scored over the retrieved context or over the answer’s own citations – decides the winner. We make four contributions. **(C1)** Answer-level citation winners are *stratum-conditional but embedder- and seed-robust*: vanilla ties GraphRAG on 1–2-hop DO-178C queries and is last at 3+–hop under all three embedders. **(C2)** GraphRAG’s walk floods the context while its synthesizer cites selectively, so scoring the retrieved set – a choice some comparisons make implicitly – ranks GraphRAG below the vanilla baseline and scoring the answer’s citations ranks it first, in all six settings, under a second traversal, and against two mitigations. **(C3)** The faithfulness decline follows *context expansion*, not the corpus: adequately powered, both Wikipedia corpora decline too, the one pipeline that expands nothing stays flat, and a hop×expansion interaction is significant. **(C4)** *Judges are unstable along separable axes*: they drift on frozen inputs, two vendors reverse which of them is stricter between corpora, and κ swings by an order of magnitude while raw agreement barely moves. The hop-wise ordering nonetheless replicates across vendors.

2 Related Work

Agentic and Adaptive RAG. Self-RAG [1] and CRAG [32] introduced reflective and corrective retrieval. Adaptive-RAG [17] routes by predicted query complexity; Probing-RAG [2] extends with internal-state probes; Search-R1 [18] trains agentic retrieval via RL; RAG-Critic [4] adds an iterative critic. None condition routing on typed-graph hop distance or expose a typed graph-lookup tool.

GraphRAG and multi-hop QA. Microsoft GraphRAG [6] and follow-ups [12, 13, 21] construct entity-relation graphs at indexing time. Han et al. [15] and GraphRAG-Bench [31] report that GraphRAG does not dominate vector RAG across all query types. Our C2 shows one reason such verdicts disagree: whether citation quality is scored over the retrieved set or over the answer’s citations decides which side wins, and C3 separates the context-flooding mechanism from its faithfulness consequence across corpora. Adjacent multi-hop benchmarks

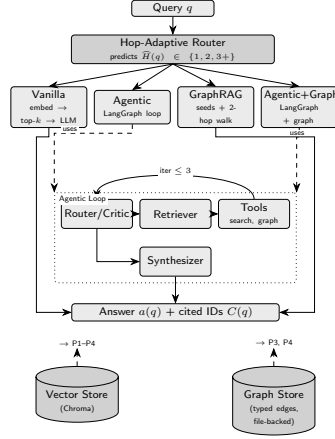


Fig. 1. Five-pipeline architecture with shared embedder, vector store, and typed-edge graph. Vanilla and GraphRAG bypass the agentic loop; Agentic and Agentic+Graph route through router/critic–retriever–tools ($\text{iter} \leq 3$) before synthesis. The adaptive pipeline (V1 rule-based or V2 learned) selects one of the four base pipelines per query.

include MuSiQue [28], MultiHop-RAG [27] and 2WikiMultihopQA [16]; we use the first and the last to replicate C1, C2 and C3 outside the requirements domain. **RAG for requirements traceability.** LiSSA [10], TVR [23], and Graph-RAG for compliance [22] evaluate single regulated domains without hop stratification or dual-judge protocols.

Citation evaluation, judge bias, and the κ paradox. ALCE [11] defined citation $P/R/F_1$ over generated answers; Wallat et al. [29] show that ALCE-style correctness is not faithfulness, motivating LLM judges. RAGChecker [25] provides a single-judge framework. Self-preference bias [30] motivates judge ensembles, and the κ paradox of Feinstein and Cicchetti [9], with Gwet’s AC1 [14], establishes the prevalence correction we apply. Our C4 adds a statistic this literature does not track: the judge’s agreement with itself under retrieval-state and date changes.

3 Method

3.1 Pipelines

All pipelines share embedder, ChromaDB vector store, a file-backed typed-edge graph, Azure GPT-5.4 generator, and grounded synthesis prompt; they differ only in retrieval. **(i) Vanilla:** dense top-10 retrieval, top-5 context, one synthesis call. **(ii) Agentic:** LangGraph [20] router-retriever-critic loop, `search_documents` tool only, iter cap 3. **(iii) GraphRAG:** vector seeds (8) + up-to-2-hop typed-graph walk (walk cap 30, context cap 15), single synthesis. This is a typed-edge local-walk retriever in the GraphRAG family, not Edge et al.’s community-summarization system [6]; we use “GraphRAG” as the family label. **(iv) Agentic+Graph:** (ii)

plus a typed-edge `graph_lookup` tool. **(v) Adaptive:** chooses among (i)–(iv) per query; we evaluate a rule-based router (V1) and a learned out-of-fold logistic regression over 11 text features and 3 query-embedding PCs (V2) that routes each predicted hop class to its empirically best pipeline.

3.2 Robustness Axes

Embedder: three models from three vendors, at three dimensionalities, so the axis is not a family variant: e5-small (384d), Azure 3-small (1536d) and BAAI bge-m3 (1024d). ChromaDB collections are rebuilt per embedder; the graph is shared. **Traversal:** the production bounded undirected walk (≤ 2 hops, cap 30) and, as a second retriever of the same family, personalized PageRank from the vector seeds – unbounded in radius, ranked by random-walk mass, all else fixed. **Corpus:** a synthetic DO-178C-style aerospace certification corpus of 1,132 requirements over 30 modules (17–76 requirements each; median 59, p90 103 tokens per requirement), generated by GPT-5.4 from a hand-authored module taxonomy and publicly released with its link annotations; and two Wikipedia paragraph-chain sets: a 200-query MuSiQue [28] subset (67/67/66 across 2/3/4-hop, mapped to our 1/2/3+-hop strata, *REFERENCES* edges only between consecutive supporting paragraphs) and a 200-query 2WikiMultihopQA [16] subset. 2Wiki strata follow reasoning structure: *comparison* is 1-hop, *compositional* and *inference* (chained) are 2-hop, *bridge_comparison* is 3+-hop. Retrieval traverses the requirements graph restricted to five traceability edge types (*derives_from*, *satisfies*, *references*, *verifies*, *refines*); system-engineering edges such as allocation are excluded as not retrieval-relevant. **Judge:** GPT-5.4, GPT-4.1 and Gemini 3.7-flash (§4).

3.3 Queries and Gold Evidence

DO-178C queries are built from the graph, not from free-form prompting: we sample directed requirement–requirement chains of length 1, 2, and 3–4 over the five whitelisted edge types and prompt GPT-5.4 to phrase one natural-language question answerable from the chain. The chain’s node set is the gold evidence, so gold size grows with the stratum (mean 2.00/3.00/4.68 IDs for 1/2/3+-hop). This yields 296 queries (100/99/97 per stratum), 85% of them cross-module; 497 of the 655 chain edges are *references*. Gold sets are therefore structural rather than human-annotated: a sampled chain is by construction a supporting path, but we do not verify that no *other* path also answers the question, so citation recall is exact while precision is conservative wherever alternative evidence exists. This affects all pipelines identically and so does not bias the comparisons.

3.4 Metrics and Judging

We report ALCE-style citation $P/R/F_1$ [11] against gold ID sets. Cited IDs are parsed from the answer text with a vocabulary-anchored parser (exact corpus IDs, longest match first); hallucinated IDs that reuse a real module prefix count

against precision, and incidental tokens (SHA-256, DO-254) are excluded. We separately report *context precision* – the gold fraction of the retrieved set handed to the synthesizer – and retrieval recall, since conflating the context set with the answer’s citations changes which architecture appears to win (§5.2). Each faithfulness judgment is a strict-JSON binary verdict over the retrieved-context block. We report Cohen’s κ , Gwet’s AC1 [9, 14], raw agreement, and McNemar exact-binomial p ; per faithfulness cell, Wilson 95% CIs and Cochran–Armitage hop-trend tests; and three same-judge controls (test–retest, embedder swap, eleven-week re-judge) on a paired 300-tuple subset (§5.4).

4 Experimental Setup

Main matrix and cross-corpus sample. DO-178C: 5 pipelines $\times$ 296 hop-stratified queries $\times$ 3 seeds = 4,440 runs under each of two embedders ($v2$, $v3$), plus 1,480 runs at one seed under the third ($v4$, bge-m3) and a 296-run personalized-PageRank arm on the Azure embedder. MuSiQue: 3 pipelines (vanilla, GraphRAG, agentic+graph) $\times$ 200 queries $\times$ 1 seed = 600 runs, Azure embedder only, with the GraphRAG arm extended to three seeds so its per-stratum n matches DO-178C’s, plus 400 runs for two mitigation variants and a 200-run GraphRAG distractor-edge control (§5.1) on the same queries and vector seeds. Agentic and adaptive are not re-run on MuSiQue (C2 needs only the three-pipeline trio). 2WikiMultihopQA: vanilla and GraphRAG $\times$ 200 queries $\times$ 3 seeds = 1,200 runs, Azure embedder.

Faithfulness judging. On DO-178C, the multi-judge protocol (GPT-5.4 + GPT-4.1) is applied to two disjoint 300-row stratified batches (60 per pipeline; seeds 42 and 43), the second judged eleven weeks after the first, yielding 1,200 paired binary judgments per embedder. Each batch is *pinned* to the same 300 (query, pipeline, repeat) tuples across embedders so C3 deltas are computed on identical items; batches are reported separately, never pooled across judging dates. On the Wikipedia corpora we judge every row with GPT-4.1 and Gemini 3.7-flash. The Gemini leg of the MuSiQue seed extension, both mitigation arms and 2Wiki ran one day after their GPT-4.1 leg, on frozen stored answer–context strings, so the pairing is row-exact; the DO-178C matrices were judged by both vendors in a single pass. Same-input controls re-judge 300 $v2$ tuples twice with GPT-5.4 (test–retest) and once eleven weeks later (temporal drift); a generator-swap control re-synthesizes 332 $v3$ rows with GPT-4.1 on frozen retrievals and dual-judges them.

Statistical protocol. 95% BCa bootstrap intervals [5] ($B=1000$, paired at the query level), Wilcoxon signed-rank with Holm correction across the 30 pipeline-pair $\times$ stratum contrasts [3, 19], and Cliff’s δ (negligible <0.147). A pipeline pair is reported as significantly different only when the BCa interval excludes zero, Holm $p < 0.05$, and $|\delta| \geq 0.147$ jointly hold. Generation is stochastic, so every query is run at three seeds; tests pool the seeds and we report the across-seed spread separately (§5.1) as a check that no reported ordering rests on a single run.

Table 1. Per-stratum citation F_1 across four settings. v2 main = DO-178C synthetic with local e5-small embeddings (4,440 runs); v3 main = same corpus with Azure text-embedding-3-small (4,440 runs); MuSiQue = 200-query Wikipedia stratified subset with Azure embedder (600 runs, vanilla / agentic-graph / graphrag only); 2Wiki = 200-query 2WikiMultihopQA subset at three seeds (1,200 runs, vanilla / graphrag only). Bold marks the per-(setting, stratum) maximum. The stratum-conditional pattern is the same on all four: vanilla is competitive at 1-hop and loses as hop distance grows.

System	v2 main (local)			v3 main (Azure)			MuSiQue (Azure)			2Wiki (Azure)		
	1h	2h	3+h	1h	2h	3+h	1h	2h	3+h	1h	2h	3+h
vanilla	.757	.712	.089	.756	.689	.195	.742	.415	.277	.990	.778	.678
agentic	.541	.414	.175	.533	.430	.203	—	—	—	—	—	—
agentic-graph	.566	.415	.219	.547	.421	.251	.457	.296	.249	—	—	—
graphrag	.751	.720	.172	.745	.707	.253	.841	.632	.463	.969	.946	.938
adaptive	.630	.499	.181	.651	.492	.254	—	—	—	—	—	—

Reproducibility. The graph store was populated before this study and carries edges beyond those annotated in the released corpus: re-loading the release with our ETL recovers 91.0% of the edges and 82.8% of the chains behind our queries. Graph-pipeline results are therefore reproducible up to that coverage; the query manifest is released with its full chains, so every gold evidence set is exact regardless. The rest of the pipeline regenerates from a single make target on locked CSVs. Azure model snapshots: `gpt-5.4-2026-03-05` (generator + judge), `gpt-4.1-2025-04-14` (judge); reasoning-mode decoding with an 8,192-token completion cap. The MuSiQue subgraph (3,996 paragraph chunks, 399 *REFERENCES* edges) is built deterministically from the `dgslibisey` MuSiQue mirror (validation split, seed 20260511), and the 2WikiMultihopQA subgraph (2,000 chunks, 332 edges) from the `voidful` mirror (seed 20260816) by the same procedure. The v2/v3 graph arms walked that original store; the bge-m3 matrix and the PPR arm walked a file-backed store built from the released link annotations, which is also what a reproduction uses (no database service needed). To check that this change of graph does not drive the comparisons, we rebuilt every v3 GraphRAG context on the file-backed store without any generation: the vector seeds are identical in 295 of 296 queries, the contexts overlap at Jaccard 0.89, and context precision (0.131 vs. 0.129) and retrieval recall (0.720 vs. 0.724) differ by at most 0.01 in every stratum. The citation side was not re-generated on the new store.

5 Results

5.1 C1: Winners are Corpus-Conditional, Embedder- and Seed-Robust

Table 1 reports per-stratum F_1 across four settings. On DO-178C, vanilla and GraphRAG are statistically tied on 1-hop and 2-hop (Wilcoxon $p=0.12/0.65$ local, negligible δ), the agentic loop costs 0.19–0.30 F_1 on those strata, and on

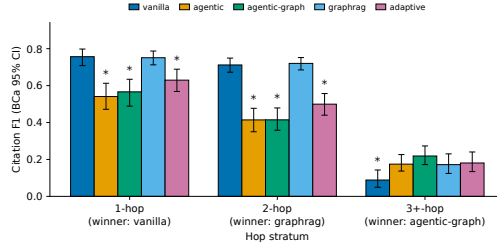


Fig. 2. Per-stratum citation F_1 on the v2 main matrix (DO-178C, local e5-small embedder): five pipelines with BCa 95% CIs; * marks pipelines significantly different from the per-stratum winner (Holm-Wilcoxon $p < 0.05$, $|\delta| \geq 0.147$).

Table 2. Per-stratum citation F_1 on DO-178C under three independent embedders (384d Microsoft, 1536d OpenAI, 1024d BAAI). The stratum-conditional ordering is stable: vanilla ties GraphRAG on 1–2-hop and is *last* at 3+-hop in all three, where a graph-aided pipeline leads.

System	e5-small (384d)			3-small (1536d)			bge-m3 (1024d)		
	1h	2h	3+h	1h	2h	3+h	1h	2h	3+h
vanilla	.757	.712	.089	.756	.689	.195	.769	.691	.114
agentic	.541	.414	.175	.533	.430	.203	.548	.404	.156
agentic-graph	.566	.415	.219	.547	.421	.251	.546	.431	.206
graphrag	.751	.720	.172	.745	.707	.253	.742	.728	.214
adaptive	.630	.499	.181	.651	.492	.254	.672	.478	.202

3+-hop the ordering inverts: vanilla drops to last and the graph-aided pipelines lead (local: agentic-graph 0.219 vs. vanilla 0.089, Holm $p < 10^{-6}$; under Azure the ordering holds but no 3+-hop pair passes the joint criterion). A third embedder from a third vendor (bge-m3) reproduces it (Table 2): vanilla is last at 3+-hop under all three, and under bge-m3 both graph arms beat it (+0.100, +0.091; Holm $p = 2.1 \times 10^{-4}$, 1.4×10^{-3}) while 1–2-hop contrasts stay negligible. On MuSiQue GraphRAG wins every stratum ($F_1 = 0.841/0.632/0.463$; $p \leq 5.1 \times 10^{-4}$, $\delta = 0.27\text{--}0.43$). Because MuSiQue’s *REFERENCES* edges connect only gold paragraphs, we re-ran GraphRAG with consecutive-distractor edges added: it loses only 0.02/0.05/0.09 F_1 and still beats vanilla everywhere ($p \leq 0.014$). On 2Wiki-MultihopQA the two are tied at 1-hop ($p = 0.15$) and GraphRAG wins 2-hop and 3+-hop by widening margins (+0.168, +0.260; $\delta = 0.48, 0.87$) – the same crossover on a graph we did not design. Seed variance is small against these effects: per-pipeline mean F_1 moves by at most 0.031 across seeds, every significant ordering holds in each seed separately, and the only rank swaps are within the vanilla–GraphRAG pairs we report as tied.

Table 3. GraphRAG floods the context and the synthesizer filters it (C2), across every setting we ran: three embedders, three corpora, and two traversals. The walk fills 11–15 slots at low precision; the answer cites 2.5–5.0 IDs at 3–4 $\times$ that precision. The ranking inverts with the measurement point in all six.

Setting	<i>retrieved context</i>		<i>answer citations</i>		
	slots	prec.	IDs	prec.	gain
DO-178C, e5-small	14.9	0.120	4.9	0.480	4.0 $\times$
DO-178C, 3-small	14.9	0.129	5.0	0.493	3.8 $\times$
DO-178C, bge-m3	14.8	0.121	4.7	0.490	4.1 $\times$
DO-178C, PPR walk	14.9	0.131	4.9	0.509	3.9 $\times$
MuSiQue	11.2	0.227	3.4	0.654	2.9 $\times$
2WikiMultihopQA	11.4	0.230	2.5	0.975	4.2 $\times$

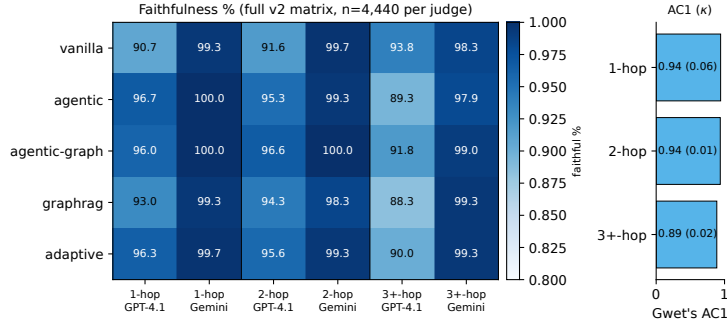


Fig. 3. Faithfulness on the full v2 main matrix (DO-178C, e5-small; 4,440 rows, ≈ 300 per cell), judged by GPT-4.1 and Gemini 3.7-flash on identical rows. *Left:* under GPT-4.1 every expanding pipeline loses 4–7 pp from 1-hop to 3+-hop while vanilla, which expands nothing, does not decline (C3); Gemini compresses every cell against the ceiling. *Right:* per-stratum Gwet’s AC1 with Cohen’s κ in parentheses – 0.89–0.94 against $\kappa \leq 0.06$ on the same verdicts, the prevalence paradox of C4. Colour scale starts at 80%.

5.2 C2: The Walk Floods the Context, the Synthesizer Filters It

Table 3 reports the same measurement in six settings: three embedders, three corpora, and two traversals. GraphRAG’s walk fills 11–15 context slots at context precision 0.12–0.23 – four to five of every six retrieved chunks are off-gold. The synthesizer then cites only 2.5–5.0 IDs per answer, at citation precision 0.48–0.98: a 2.9–4.2 $\times$ precision enrichment over its own context, in every setting. GraphRAG’s F_1 edge is therefore a *recall* effect: the walk surfaces gold chunks dense retrieval misses, and the synthesizer declines to cite most of the noise that rides along. Evaluations that score the retrieved set as the attribution set – as some GraphRAG comparisons do – measure the flooding and miss the filtering, and invert the ranking: by context precision GraphRAG sits below the vanilla baseline in every setting (0.12–0.23 against 0.27–0.37), while by answer-citation F_1

it is the highest-scoring pipeline in every setting (0.551, 0.571, 0.565, 0.585, 0.646, 0.951 in the row order of Table 3). The two measurements do not merely differ in magnitude; they disagree on the sign of the comparison that the architecture debate turns on.

Two controls show the split is not an artifact of our own design choices. First, *traversal*: replacing the bounded 2-hop walk with personalized PageRank over the same graph – unbounded in radius and ranked by random-walk mass rather than hop distance, with embedder, seeds and context cap held fixed – changes almost nothing (context precision 0.131 vs. 0.129, and 0.131 for the 2-hop walk rebuilt on the same file-backed graph, §4; enrichment $3.9\times$ vs. $3.8\times$; citation $F_1 +0.016$ at Cliff’s $\delta=0.036$, which our joint criterion rejects as negligible). Second, *mitigation*: neither obvious fix helps. Reranking the walked neighbours by query similarity and keeping the top five leaves everything where it was (context precision $0.227\rightarrow 0.232$, citation $F_1 +0.010$, $p=0.34$), while tightening the traversal budget actively hurts (walk $30\rightarrow 10$, context $15\rightarrow 8$: citation $F_1 -0.126$, $p=2.6\times 10^{-11}$, and context precision *falls* to 0.194). This is what C2 predicts: the synthesizer already performs the filtering, so filtering again upstream is redundant, and cutting the budget only removes the recall the advantage rests on.

5.3 C3: The Decline Follows Context Expansion, Not the Corpus

Judged on 300-row subsets, the faithfulness decline looks *corpus-conditional*: present on DO-178C, absent on Wikipedia chains. That reading does not survive adequate power. The MuSiQue arm had 67 rows per stratum; its 1-hop to 3+-hop drop (9.2 pp) was larger than the 4.7 pp that reaches significance on DO-178C, but detecting it at 80% power needs about 210 rows per cell. At three seeds ($n=600$) the decline is significant under both judges: 0.930/0.876/0.823 under GPT-4.1 ($p=1.1\times 10^{-3}$) and 0.955/0.945/0.869 under Gemini ($p=1.2\times 10^{-3}$). 2WikiMultihopQA, judged in full ($n=1,200$), declines as well ($p=5.0\times 10^{-2}$ and 4.8×10^{-6}), so the decline replicates on three corpora under two vendors.

What separates pipelines is whether they expand the context at all. Vanilla, the only pipeline that retrieves a fixed top-5 and expands nothing, shows no significant 1-hop to 3+-hop drop under GPT-4.1: -1.5 pp on DO-178C ($n=2,072$, per-embedder $p=0.16, 0.99, 0.76$) and $+1.1$ pp on 2Wiki ($p=0.66$), against 6.3 and 6.7 pp for the four expanding pipelines. A logistic fit of faithfulness on hop, an expansion indicator and their interaction makes this a single test: the interaction is negative and significant on the pooled DO-178C matrix ($b=-0.599$, $p=1.0\times 10^{-7}$, $n=10,360$) and negative but not significant on 2Wiki ($b=-0.286$, $p=0.27$), whose arm is a twentieth the size. Under Gemini the ordering holds but the test does not resolve: near the ceiling the baseline itself acquires a slope ($b=-0.729$, $p=0.012$) and the interaction is not separable from it ($p=0.11$), though expanding arms still drop further than vanilla (2.8 vs. 1.5 pp on DO-178C, 12.7 vs. 7.7 pp on 2Wiki) – a lenient judge preserves the ranking and loses the resolution (C4). We do not claim every expanding pipeline declines in every cell: GraphRAG is marginal under bge-m3 ($p=0.065$), and the agentic arms decline in four of six

Table 4. Judge stability. *Top*: same judge, frozen input — the answer and context strings are identical and only the judging date differs, so the fall from 0.76 to 0.14 isolates drift from any retrieval effect. *Middle*: same judge, input changed by embedder swap. *Bottom*: cross-vendor agreement on identical rows. Which judge is stricter depends on the corpus — Gemini is markedly more lenient on DO-178C, indistinguishable on the capped MuSiQue arm, and stricter on 2Wiki. Raw agreement and AC1 move together across the eight settings while κ swings by a factor of fifteen, ordered by distance from the ceiling.

<i>Same judge, frozen input (answer + context identical)</i>				
GPT-5.4, re-judged same day				$\kappa = 0.764$, raw agr. 0.88
GPT-5.4, re-judged 11 wk later				$\kappa = 0.138$, raw agr. 0.56
GPT-4.1, re-judged 11 wk later	$\kappa = -0.05 / -0.00$	(v2/v3), raw agr. 0.48 / 0.51		
GPT-4.1, full matrix vs. May				$\kappa = -0.046$, raw agr. 0.48, +43 pp
<i>Same judge, input changed by embedder swap</i>				
GPT-5.4 (v2 vs v3)				$\kappa = 0.137$, raw agr. 0.59
GPT-4.1 (v2 vs v3)				$\kappa = 0.480$, raw agr. 0.74
<i>Cross-vendor, same rows, one judging window (Gemini 3.7-flash $\times$ GPT-4.1)</i>				
<i>Setting</i>	raw	κ	AC1	faithful (G / 4.1)
DO-178C e5-small ($n=4,440$)	0.929	0.029	0.923	0.993 / 0.933
DO-178C 3-small ($n=4,440$)	0.913	0.125	0.903	0.980 / 0.917
DO-178C bge-m3 ($n=1,480$)	0.939	0.047	0.935	0.991 / 0.945
DO-178C PPR walk ($n=296$)	0.926	0.058	0.919	0.983 / 0.936
MuSiQue, 3 seeds ($n=600$)	0.850	0.172	0.817	0.923 / 0.877
MuSiQue rerank ($n=200$)	0.820	0.171	0.772	0.935 / 0.825
MuSiQue budget cap ($n=200$)	0.875	0.419	0.841	0.875 / 0.880
2WikiMultihopQA ($n=1,200$)	0.899	0.435	0.877	0.888 / 0.914

embedder–pipeline combinations. The GPT-5.4 leg has not been re-run at this power.

5.4 C4: Judges Are Unstable on Frozen Inputs and on Changed Ones

Table 4 separates judge instability measured on *frozen* inputs, where no retrieval is re-run, from instability under a changed retrieval state. *Frozen inputs*. Re-judging the identical 300 stored (answer, context) pairs gives GPT-5.4 $\kappa=0.76$ (raw agreement 0.88) on the same day but only $\kappa=0.14$ (0.56) eleven weeks later, a +15 pp leniency shift that rejects a stationary judge (binomial $p<10^{-44}$). GPT-4.1 on the same tuples falls to chance agreement with its own earlier verdicts ($\kappa=-0.05/-0.00$), +41 pp more lenient. With inputs fixed, this is judge drift alone. *Changed retrieval state*. After an embedder swap, which changes contexts and answers on 94% of tuples, GPT-5.4 flips 41% of verdicts ($\kappa=0.137$) and GPT-4.1 26% ($\kappa=0.48$): the judge more stable under embedder swap is the less stable across time. *Across vendors*. On identical rows, Gemini 3.7-flash is far

more lenient than GPT-4.1 on all four DO-178C settings (0.98–0.99 vs. 0.92–0.95 faithful; McNemar $p < 10^{-9}$), indistinguishable on the budget-capped MuSiQue arm (0.875 vs. 0.880, $p = 1.00$), and *stricter* on 2WikiMultihopQA (0.888 vs. 0.914, $p = 6.2 \times 10^{-3}$). Strictness is a property of the judge–corpus pair and cannot be calibrated once and reused. The same eight settings expose the prevalence paradox [9] within one judge pair: raw agreement moves only from 0.82 to 0.94 and Gwet’s AC1 tracks it (0.77–0.94), but κ swings from 0.03 to 0.44, ordered almost exactly by distance from the ceiling. On 2Wiki the pair agrees *less* often than on DO-178C (0.899 vs. 0.93) yet appears fifteen times more reliable by κ (0.435 vs. 0.029). Leniency and *trend* nonetheless come apart: near the ceiling, Gemini still reproduces the hop-wise decline on both DO-178C matrices ($p = 4.3 \times 10^{-3}$, $p < 10^{-16}$) and on both Wikipedia corpora. The ordering a judge induces replicates across vendors; its strictness does not.

Between judges, and why we re-judged everything. Inter-judge κ halves under embedder swap on identical tuples (0.30 $\rightarrow$ 0.17; McNemar $p < 0.001$), and on the 300-row MuSiQue batch GPT-5.4 and GPT-4.1 agreed at chance (raw agreement 0.48, AC1 0.08 – genuine disagreement, not a prevalence artifact). Verdicts this unstable cannot carry a cross-corpus claim, which is why the full matrix was re-judged inside a single window (§5.3); no faithfulness comparison we report spans judging dates.

6 Discussion

Why does expansion cost faithfulness? GraphRAG’s traversal expands candidates well beyond what dense retrieval surfaces, and the synthesizer filters most of the noise out of its citations (C2). What it cannot filter is the influence of that noise on the answer itself: faithfulness is judged against the full retrieved block, so every chunk the walk adds is a chunk the answer can be found unfaithful to. That mechanism is indifferent to what the edges connect, which is why the decline appears on typed DO-178C traceability links and on Wikipedia paragraph adjacency alike, and why the one pipeline that adds nothing is the one that stays flat. The cost scales with the number of expanded chunks rather than with their semantic distance — consistent with C2’s finding that reranking those chunks for relevance changes nothing.

Implications for LLM-judge practice. Because judges drift with the retrieval state and with the date (§5.4), a single-judge comparison across retrieval modules reports embedder- and date-conditional verdicts, not invariant faithfulness, and multi-judge ensembles help only partially since inter-judge agreement drifts too. Paired-tuple judging under a fixed embedder, inside one judging window, is the least that is needed for comparable faithfulness numbers.

Threats to validity. Construct. ALCE-style F_1 does not measure rationale quality; we pair it with multi-judge faithfulness, while acknowledging LLM-judge bias [29, 30]; C4 quantifies this directly. A third judge from a different vendor

(Gemini 3.7-flash) now covers every matrix in the study, and C4 shows its strictness relative to GPT-4.1 reverses between corpora. *External.* The DO-178C-style corpus is a single synthetic aerospace dataset, generated by the same model family that answers and judges; absolute F_1 levels characterize this single regulated-domain setting and may not transfer. A generator-swap control (GPT-4.1 re-synthesizing 332 rows on frozen retrievals) reproduces the architecture ordering with slightly higher F_1 , and its answers show the same hop-wise faithfulness decline under both judges, so neither result is GPT-5.4-specific; the corpus-author leg of the circularity remains. The two Wikipedia corpora are the main external-validity check: both are judged in full by two vendors and both replicate C1’s crossover, though only DO-178C carries all five pipelines. Our graph pipeline is one typed-edge walk implementation, though swapping it for personalized PageRank leaves C2 intact; a full third-party GraphRAG system may still flood or filter differently. *Mitigations.* The two unsupervised fixes we tried, reranking and a tighter budget, fail for the reason C2 gives (§5.2); a learned, question-conditioned evidence selector remains open. *Statistical.* Per-stratum n for citation metrics is 97–100 on DO-178C and 66–67 on the Wikipedia corpora (paired at the query level), the lower edge of BCa stability under heavy skew; a percentile fallback shows no qualitative reversal. Faithfulness cells now carry $n \approx 300$ after full-matrix judging, which is what overturned the corpus-conditional reading in C3. The hop $\times$ expansion interaction is significant on DO-178C but only same-signed on 2Wiki, whose arm is a twentieth the size; we report it as directional there.

Artifact availability. The corpus with its link annotations, the query manifest with full chains, the code, and every scored and judged CSV are released under CC-BY-4.0 (links withheld for review); the Wikipedia subsets are rebuilt by seeded scripts.

7 Conclusion

Around a fixed five-pipeline matrix, varying embedder, corpus, traversal and judge shows that moving the citation-measurement point from context to answer inverts the architecture ranking; that the faithfulness decline follows context expansion rather than the corpus; that stratum winners are embedder- and seed-robust; and that judges drift on frozen inputs while κ hides it. Robustness on these axes, and an explicit choice of measurement point, is the minimum bar for RAG architecture claims.

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